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Reservoir Souring Modeling for Low Salinity Waterflood in a Deepwater Field

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Abstract

Low salinity (LoSal) water injection has recently started in a deepwater field to enhance oil recovery. There is a need to assess the reservoir souring risks arising from LoSal plant performance excursion beyond the design sulphate specification for the LoSal water. Thus, detailed reservoir souring modeling work is required to quantify the impact of LoSal plant performance excursion on the prediction of H₂S production profile – the basis for reservoir souring risk assessment.

This paper presents detailed reservoir souring modeling work based on one-way coupling of numerical reservoir simulator with commercially available reservoir souring simulator. In this approach, the results of reservoir simulation, including flux arrays for all the flowing phases, are fed into the reservoir souring model to simulate the biological reduction of sulphate and generation of H₂S as well as H₂S transport and partitioning in subsurface and beyond. Both subsurface uncertainty and LoSal plant excursion scenarios are considered as part of the sensitivity analysis to define the range of H₂S production profiles, which enables comprehensive assessment of reservoir souring risks.

The results of the comprehensive reservoir souring modeling in this paper demonstrate clearly that even though the reservoir in this deepwater field is relatively hot, and the LoSal waterflood includes robust sulphate removal, the combination of low injection water temperatures, large volume of injected LoSal water, as well as LoSal plant sulphate level excursion still causes substantial reservoir souring risks in certain scenarios. In particular, the amount of LoSal water injected, in combination with the magnitude of LoSal plant performance excursion, would significantly affect the H₂S production profile. Larger amounts of LoSal water being injected tend to generate larger cool zones around the injectors, which in turn cause more aggressive biomass (sulphate reducing bacteria) growth near the injectors, and therefore more rapid H₂S generation. These insights are important, as in deepwater reservoirs with stringent export H₂S specifications, even LoSal water injection would pose significant reservoir souring risks. These insights provide not only robust basis for souring risk assessment, but also options for mitigating the risks, such as barriers (scavenging) and reservoir / production management strategies.

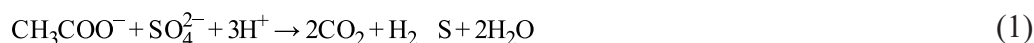
In summary, the work documented in this paper represents the first attempt to simulate the reservoir souring process for LoSal waterflood in a deepwater reservoir. It highlights the reservoir souring risks even for an environment considered unfavorable for bacterial sulphate reduction (e.g., lower sulphate availability

due to LoSal water, higher reservoir temperatures), with unique insights gained for understanding reservoir souring mechanisms.

Background and Context

Reservoir souring refers to the phenomenon of originally sweet reservoirs without indigenous H₂S becoming sour (i.e., producing H₂S) as a result of water injection. It is generally accepted that the sulphate being supplied by the injection water, together with the carbon sources either available in-situ, or being supplied by the injection water, or both, constitutes the necessary conditions for Biological Sulphate Reduction (BSR) to take place. If conditions are favorable in-situ (e.g., sufficiently low temperature, the absence of very high pressure and hypersaline environment, etc.), biomass growth will occur, which will lead to biological reduction of sulphate and generation of H₂S. Injection water can have many different sources. In deepwater offshore, it is common to use seawater as injection water, which is rich in sulphate. Sometimes seawater injection is complemented by produced water reinjection, which is typically rich in carbon sources (e.g., Volatile Fatty Acids, or VFA) for biological consumption.

BSR, the mechanism responsible for reservoir souring, can be conceptually described by the following relationship (Eq. 1), with VFA (Acetate) as an example of carbon source (Haghshenas et al., 2012):



Reservoir souring presents a significant challenge to hydrocarbon production with waterflood, often resulting in production deferral, risks to personnel health and safety, corrosion / integrity risks and potential product value discount. These negative consequences are well documented in the open literature (Al-Rasheedi et al., 1999; Khatib and Salanitro, 1997; Vance and Trasher, 2005; Ziaei, et al., 2013).

The deepwater field investigated in this study has been on production since the early 2000s. However, water production has only started very recently, with LoSal water injection for enhanced oil recovery, and no produced water re-injection. Only the producers in the southwestern part of the field are supported by the LoSal injection. The producers in the Northeastern part of the field are under primary depletion only. There is no evidence that these two parts of the field are hydraulically connected. The topsides where the LoSal plant is located are the focus of this study. The reservoirs in this deepwater field are Miocene sandstone stacked-pay reservoirs. The reservoir temperatures range from above 80°C in the crest to close to 120°C towards the oil water contact. So the reservoir temperature range itself is not conducive to reservoir souring, as the in-situ reservoir temperature exceeds the commonly accepted threshold of 80°C for meaningful BSR to take place. Reservoir pressure is in excess of 10,000 psia, even in areas where LoSal water injection is meant to target. As a result of high reservoir pressure, the in-situ oil is highly undersaturated. Hence, there is no free gas in subsurface, regardless of the subsurface scenarios. Some views expressed in the open literature argue that in-situ pressure well above 7,000 psia is inhibitive to BSR (Mota et al., 2013; Picard and Daniel, 2013; Yayanos, 2019). Therefore, with all these factors combined (low sulphate supply, high reservoir pressure and temperature, no produced water re-injection), one would have the impression that the reservoir souring risks for this field should be low.

It is difficult to conclude that those views on the impact of high reservoir pressure are fully supported by irrefutable laboratory and field evidence / experience. There are examples in the open literature that take a more cautious approach when it comes to assessing reservoir souring risks under high reservoir pressure conditions (Izgec, 2023). In this study, a similarly conservative approach is taken, i.e., any inhibitive effect of high reservoir pressure is not considered. From the risk management perspective for this field, such conservative approach is warranted, as there are stringent specifications for topsides gas export H₂S concentration strictly enforced by the third-party owner / operator of export lines.

To properly assess the reservoir souring risks, a one-way coupled commercially available reservoir souring modeling tool has been used. In this approach, the reservoir dynamic simulation results at each

time step, including flux arrays for all flowing phases, are fed into the reservoir souring model, which then simulates the biomass growth, H_2S generation, transport and partitioning in subsurface, as well as in production systems during the same time step. The reservoir souring modeling results are not fed back to reservoir simulation model, thus having no effect on the multiphase flow in subsurface modelled in the dynamic reservoir model. Hence, the coupling between dynamic reservoir simulation and reservoir souring simulation is one-way. The workflow of such one-way coupling between reservoir simulation and reservoir souring modeling is illustrated in Fig. 1.

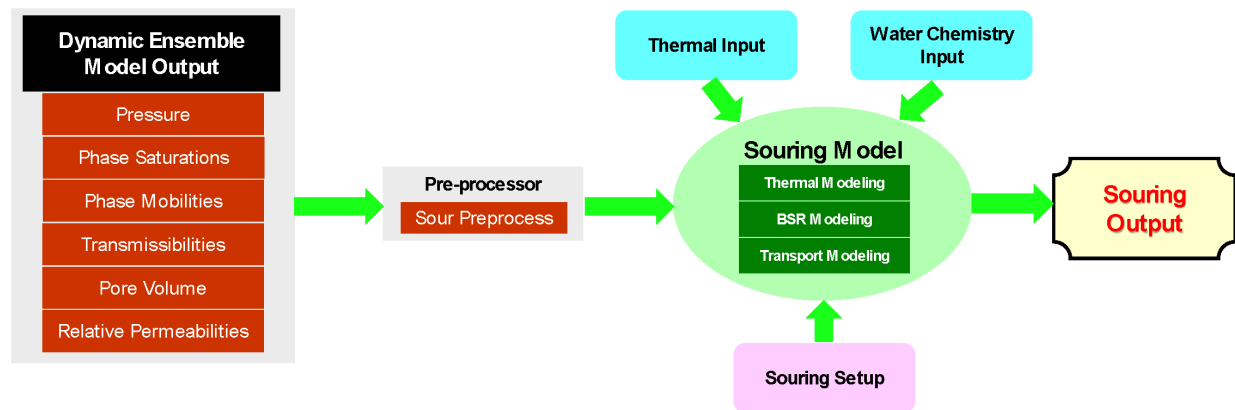


Figure 1—One way coupled workflow between dynamic reservoir simulation model and reservoir souring model

One could argue that in some instances, biomass growth could plausibly be locally substantial, potentially impacting injectivity. The one-way coupling approach essentially implies that such impact on injectivity is ignored. Since reservoir souring modeling tool has become a mainstay tool for many of the operators to assess reservoir souring risks, this one-way coupling approach has been widely adopted.

Reservoir Dynamic Modeling Basis

A full-field reservoir simulation model, with close to two million cells (259*209*36), has been built, covering a wide range of subsurface scenarios. Hundreds of ensemble realizations have been constructed to match the production and pressure history. Probabilistic distribution based on cumulative oil production has been established from these ensemble realizations. Out of these ensemble realizations, three are chosen to represent a range of subsurface scenarios for assessing reservoir souring risks, based on water injection and production behavior – the so called "Low", "Reference" and "High" scenarios. It is important to note that for modeling reservoir souring in such highly undersaturated reservoirs, the behavior of water injection and production is the primary control for the severity of reservoir souring – therefore it should be the basis for selecting ensemble scenarios underpinning reservoir souring modeling. Fig. 2 shows these three scenarios for reservoir souring modeling, along with the distribution of the entire ensemble scenarios.

According to Fig. 2, the "Low" reservoir souring scenario corresponds roughly the P90 realization of the ensemble scenarios. In a probabilistic sense, the likelihood of the "Low" reservoir souring scenario happening is high. In addition, the so called "Reference" reservoir souring scenario corresponds roughly to the P10 realization of the ensemble scenarios, or the likelihood of the "Reference" scenario happening is low. Finally, the so called "High" reservoir souring scenario lies between P50 and P10, meaning it is somewhat likely to happen.

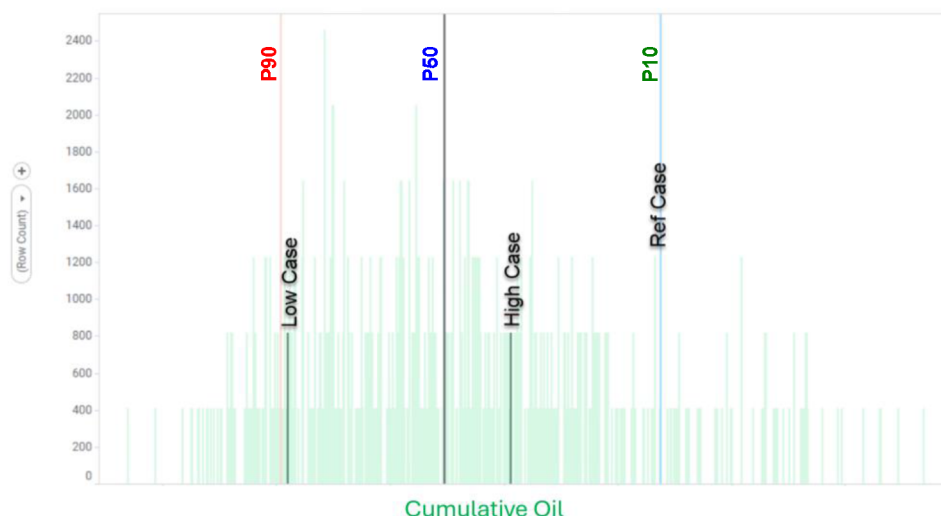


Figure 2—Three scenarios for reservoir souring modeling, among the entire dynamic ensemble modeling scenarios

Fig. 3 shows the field instantaneous and cumulative water injection profiles for the three scenarios. It is self-evident that the Low scenario shown in Fig. 3 shows the lowest cumulative water injection (red curve), hence its name. Likewise, the High scenario shown in Fig. 3 shows the highest cumulative water injection (blue curve). In all of these three scenarios, cumulative water injection constraints are imposed, based on benchmark studies. In other words, any injector will be shut in if its cumulative water injection volume exceeds the benchmarked constraint. Fig. 4 shows both the cumulative water production and the water cut for these three scenarios. Similar to the water injection profiles shown in Fig. 3, the Low scenario shown in Fig. 4 corresponds to the lowest cumulative water production, as well as the lowest water cut. The High scenario shown in Fig. 4 corresponds to the highest cumulative water production, as well as the highest water cut.

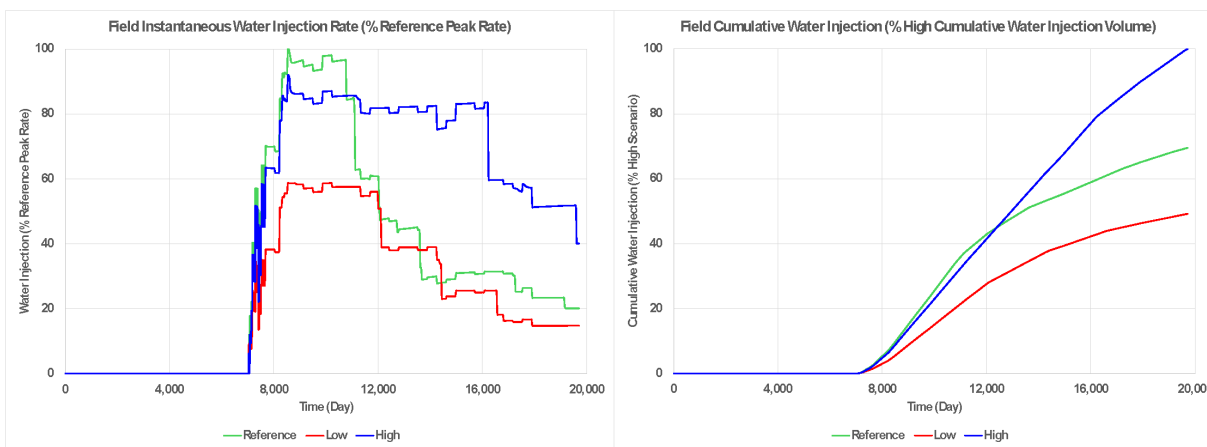


Figure 3—Field instantaneous and cumulative water injection profiles for the three reservoir souring scenarios

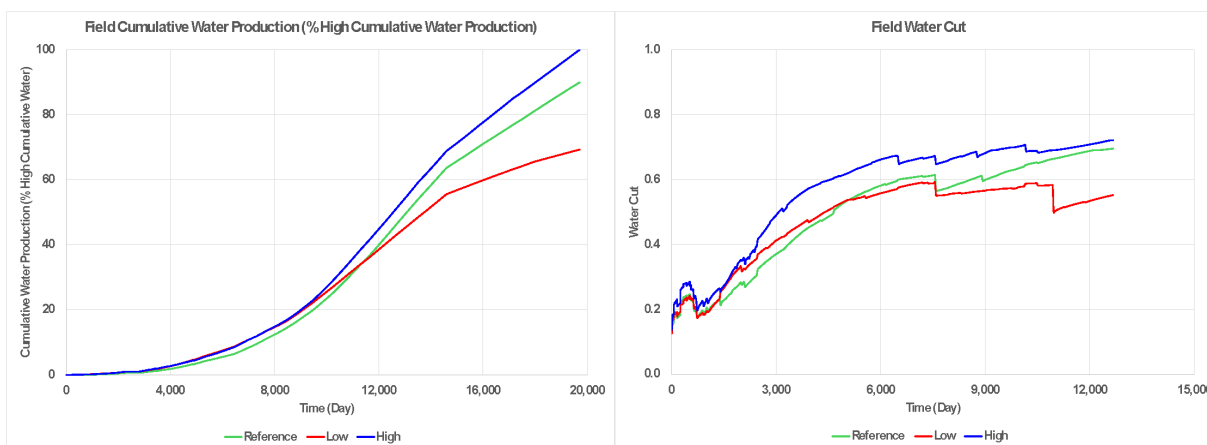


Figure 4—Field cumulative water production and water cut profiles for the three reservoir souring scenarios

Fig. 5 shows the instantaneous gas production profile for the three scenarios. While there is no free gas in subsurface, the solution gas will evolve at surface, taking H_2S into the export gas stream because of H_2S preferential partitioning into vapor phase under surface conditions. Therefore for assessing risks of gas export specification violation, gas production profiles for the three scenarios are relevant – the vapor phase H_2S concentration (in ppmv) is inversely proportional to the gas volume. Finally, it is important to point out that there is one inherent inconsistency in the one-way coupling treatment for this study – the reservoir dynamic simulation ensemble models are all isothermal, including the three scenarios chosen for reservoir souring modeling, while the reservoir souring models built from these three cases are thermal models. In fact thermal effect is among the most important controlling factors for reservoir souring (Alkan et al., 2024), particularly in this field with relatively high in-situ formation temperature. The injection of LoSal water will create a cool zone around the injectors, increasing fluid viscosity near the wellbore and resulting in reduced injectivity. Such effect is not captured in the isothermal reservoir dynamic simulation ensemble models. However, based on internal mechanistic reservoir souring modeling, which takes such effect into account, the impact on reservoir souring prediction appears to be secondary, and only visible in the long-term prediction (Amer, 2025). As a result, such inconsistency is accepted as a reasonable approximation in this study. It should also be noted that the reservoir souring models for the chosen three ensemble scenarios are fully thermal, therefore accounting for the effect of cool zone on biomass growth and H_2S generation.

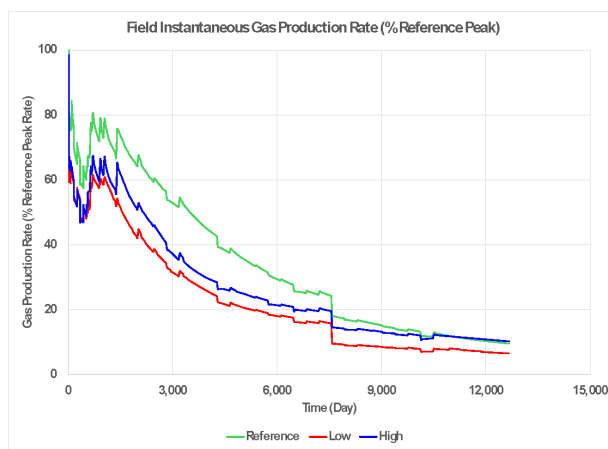


Figure 5—Field instantaneous gas production profiles for the three reservoir souring scenarios

Reservoir Souring Modeling - Input

Thermal Modeling Input

The reservoir souring thermal modeling input discussed below applies to all three reservoir souring scenarios.

Rock and Fluid Thermal Properties. For rock thermal properties, typical sandstone heat capacity and thermal conductivity values in reservoir souring default setting are used for reservoir rocks. In reservoir souring thermal model, default heat capacity and thermal conductivity values for shale are assumed for both overburden and underburden rocks. All the thermal property values are assumed to be constant in the default setting. For fluids, the reservoir souring default heat capacity and thermal conductivity values for oil, gas and water are used. Similar to rock thermal property values, thermal properties for fluids are also assumed to be constant. [Table 1](#) summarizes the rock and fluid thermal properties used for all three reservoir souring scenarios.

Table 1—Reservoir souring default rock and fluid thermal property input for all three reservoir souring scenarios

Rock Type	Volumetric Heat Capacity [kJ/(m ³ °C)]	Thermal Conductivity [W/(m°K)]
Sandstone	2400.0	2.755
Shale	2128.5	1.687
Fluid Type	Heat Capacity [kJ/(kg°K)]	Thermal Conductivity [W/(m°K)]
Oil	2.30	0.133
Gas	1.50	0.021
Water	4.19	0.619

Temperature distribution for Initialization. For temperature spatial distribution in reservoir souring thermal model initialization, a geothermal gradient model with a range of 0.0125 °C/m to 0.0172 °C/m is applied.

LoSal Injection Water Arrival Temperature at Sand-face. The lowest LoSal injection water arrival temperature at sand-face, 75°F (or 23.9°C), is used for reservoir souring thermal modeling. It is worth noting that in a separate internal study ([Amer, 2025](#)), reservoir souring thermal modeling is shown to benchmark very well with other well-established fully thermal reservoir modeling tools.

Souring Options

For H₂S generation, six options are available in reservoir souring model. Consistent with the conservatism discussed above for reservoir souring risk assessment, both the "Medium H₂S Generation (New)" and the "High H₂S Generation (New)" options are evaluated. Both options give similar predictions. In all the reservoir souring modeling results discussed in this study, the "Medium H₂S Generation (New)" option is chosen.

It is assumed that the reservoirs in this field do not have H₂S scavenging capacity – again consistent with the conservative approach taken in this study. For H₂S partitioning coefficients, two tables are generated specifically for this study, using the Equation of State fluid characterization for the field, and the H₂S module in a commercially available PVT modeling package – one for downhole conditions and another for surface conditions. Both of these tables are a function of salinity (or TDS). [Fig. 6](#) shows an example for the surface condition H₂S partitioning coefficient table.

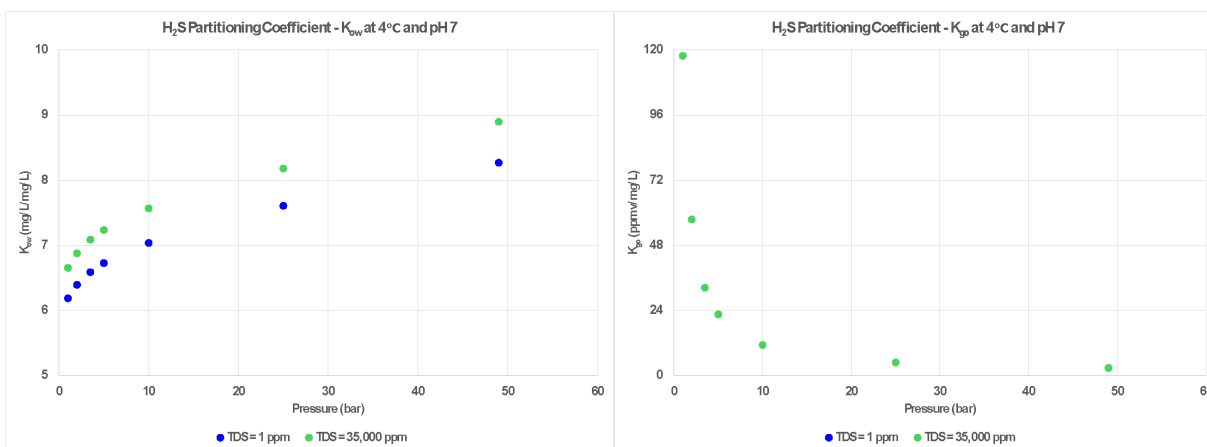


Figure 6—H₂S partitioning coefficients as a function of pressure and salinity under surface conditions – an example

Note that in reservoir souring modeling, the default H₂S partitioning tables are not a function of salinity. But for this study in which the injected LoSal water has a substantially lower TDS than that of the formation water, the need to generate H₂S partitioning tables as a function of TDS is self-evident. No preventive and mitigation measures (such as biocide treatment, nitrate injection and H₂S scavenger injection) are considered in the reservoir souring modeling. Gas lift is modeled in reservoir souring modeling, as intended in the reservoir dynamic modeling ensemble runs.

Water Chemistry Input

Formation Water. Based on water chemistry data, formation water, including aquifer water, is initialized with 57 mg/L of sulphate, 296 mg/L of Volatile Fatty Acid (VFA), 50 mg/L of phosphorus, 50 mg/L of nitrogen with a TDS of 28,317 mg/L and a pH value of 6.3. No coupled aqueous geochemistry modeling is considered, so pH is treated as passive tracer, similar to how TDS is treated. Formation water is also initialized with varying but low levels of H₂S (<0.16 mg/L), as a few of the producers in this field have been producing low levels of H₂S prior to LoSal water injection. Benzene and Toluene are included as additional carbon sources, with oil biodegradation option disabled. These formation water input parameters apply to all three reservoir souring scenarios.

LoSal Injection Water. The baseline LoSal injection water contains 50 mg/L of phosphorus, 50 mg/L of nitrogen with an average TDS of 1,800 mg/L and a pH value of 6.9, but no VFA and no H₂S. This baseline LoSal water chemistry input applies to all three reservoir souring scenarios. In addition to the baseline input, the sulphate level reflecting the LoSal plant performance is incorporated, as shown in Fig. 7. Furthermore, a large number of LoSal water sulphate level sensitivities are considered for the range of LoSal plant performance excursion scenarios in the future. The LoSal water sulphate level sensitivities investigated in this study range from 0 mg/L to 5X the BoD value.

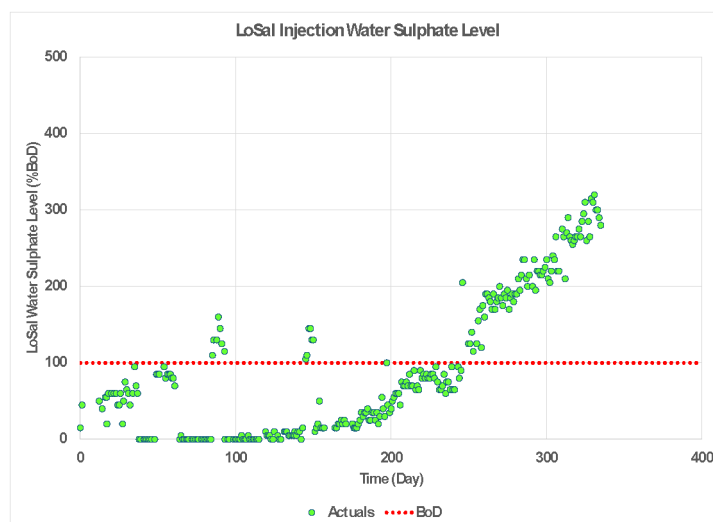


Figure 7—LoSal injection water sulphate levels since the start of water injection

Reservoir Souring Modeling – Prediction

Short Term Prediction

Short term reservoir souring modeling prediction is needed to underpin the immediate business / operational decisions to improve LoSal plant performance, as well as to assess the risks of gas export H_2S specification violation. The short-term prediction covers an approximately 3-year period from the start of LoSal water injection. Fine time steps (3-day time steps) are used in the reservoir souring modeling runs for all three reservoir souring scenarios for prediction. For history (i.e., from the start of LoSal water injection to present, even finer, daily time steps are taken in reservoir souring modeling. Sensitivity runs have been made on 3X and 5X BoD values of LoSal water sulphate concentrations for the three reservoir souring scenarios. The two sulphate excursion sensitivities are applied to the remainder of the 3-year period, after honoring the LoSal water sulphate history in the first year of LoSal water injection, as illustrated in Fig. 7.

Long Term Prediction

Long term reservoir souring modeling prediction is primarily for assessing the risks of gas export H_2S specification violation. The long- term reservoir souring modeling prediction results can also be used to assess the material integrity risks of wells / completion and subsea flowlines. The long-term reservoir souring prediction covers an approximately 35-year period from the start of LoSal water injection. A large number of LoSal water sulphate level sensitivities are run to assess the risks of gas export H_2S specification violation and provide guidance to long-term management strategy for LoSal plant sulphate level excursion. Due to the concern around runtime, coarser time steps (monthly) are used. The comparison of finer and coarser time steps done for the short-term prediction shows that monthly time steps are numerically reasonable approximation to the fine time steps of 3 days.

Results and Discussion

Short Term Prediction

Fig 8 shows the short-term prediction topsides gas H_2S concentration profile for the three reservoir souring scenarios, with LoSal water sulphate level sensitivity cases of 3X and 5X BoD values. It is very clear from Fig. 8 that for all the three reservoir souring scenarios with aggressive LoSal water sulphate level excursion, the topsides are in no danger of exceeding the gas export H_2S limit in the short term (i.e., within ~3 years of the start of LoSal water injection). It is also clear from Fig. 8 that the reservoir souring prediction results for

topsides gas H_2S concentrations are not dependent on the magnitude of LoSal water sulphate level excursion, regardless of the reservoir souring scenario. Fig. 9 shows the comparison of prediction of topsides gas H_2S concentrations between fine time steps (3-day) and course time steps (monthly) for the Reference scenario with 3X BoD value for LoSal water sulphate level excursion. It is evident from Fig. 9 that the prediction results from fine and course time step runs are in reasonable agreement. As a result, it is justified to use course time steps (monthly) for long term reservoir souring predictions. There are fundamental reasons why in the short term, violation of gas export H_2S limit is very unlikely, as illustrated in Fig. 8. To understand those reasons, it is useful to examine the behavior of water injection / production indicators (passive tracers). One perfect example of such indicators (passive tracers) is VFA, as there is a sharp contrast between VFA levels in the formation and LoSal injection water. Fig. 10 shows the VFA profiles of the Reference scenario for the formation water, injected LoSal water of both 3X BoD and 5X BoD sensitivities, and produced water for both 3X BoD and 5X BoD sensitivities. It is evident from Fig. 10 that the produced water has almost identical VFA concentration as that of the formation water, indicating the lack of injection water breakthrough during the short-term prediction period. This observation holds for all three reservoir souring scenarios, with both LoSal sulphate level excursion sensitivities.

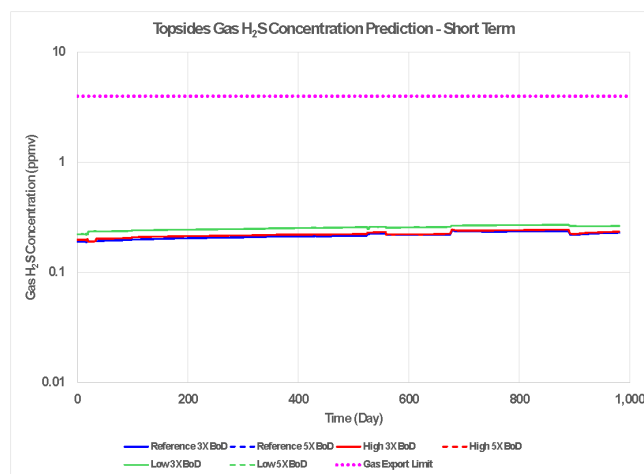


Figure 8—Topsides gas H_2S profiles for the short-term prediction

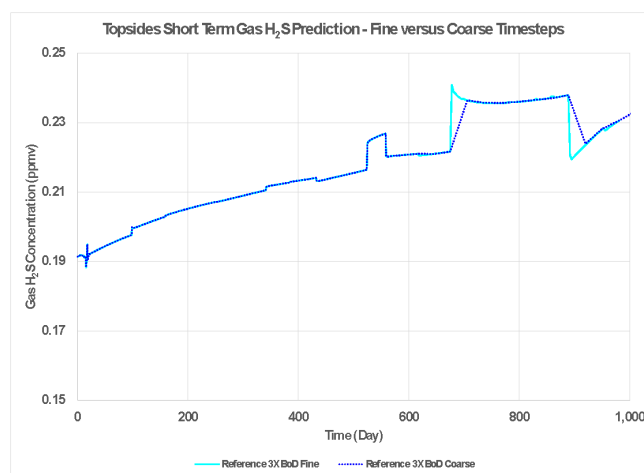


Figure 9—Comparison of fine and course time steps for the Reference scenario runs

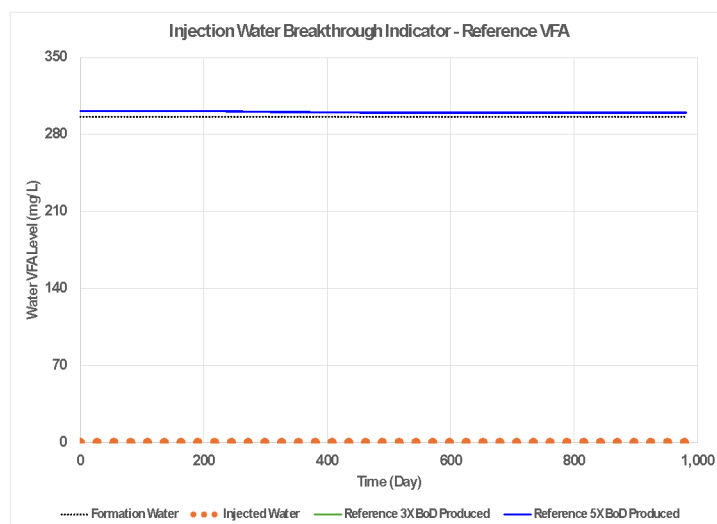


Figure 10—VFA profiles for formation, LoSal injection and produced water – Reference case with 3X and 5X BoD sulphate levels

Another angle through which one can examine the behavior of VFA as a passive tracer is a temporal map comparison of VFA distribution in the reservoir. Figs. 11 and 12 show such comparison for the Reference scenario with 5X BoD LoSal water sulphate level. It is clear from such comparison that the low VFA levels representing the injected LoSal water are all still in the immediate vicinity of the injectors at the end of the short-term prediction. In other words, injection water breakthrough has not yet taken place at the end of the short-term prediction. Since there is no free gas in the reservoir (even for the entire field life), no injected water breakthrough means no BSR generated H_2S would be produced. The low levels of H_2S being produced, as shown in Figs. 8 and 9 are entirely due to the indigenous H_2S already present in this reservoir prior to LoSal water injection. Such map comparison also holds for all three reservoir souring scenarios, with both LoSal sulphate level excursion sensitivities. In essence, during the short-term prediction period, there has not been sufficient amount of sulphate supplied to the reservoir for BSR to take off, regardless of LoSal water excursion magnitude.



Figure 11—Initial VFA distribution at the start of LoSal water injection

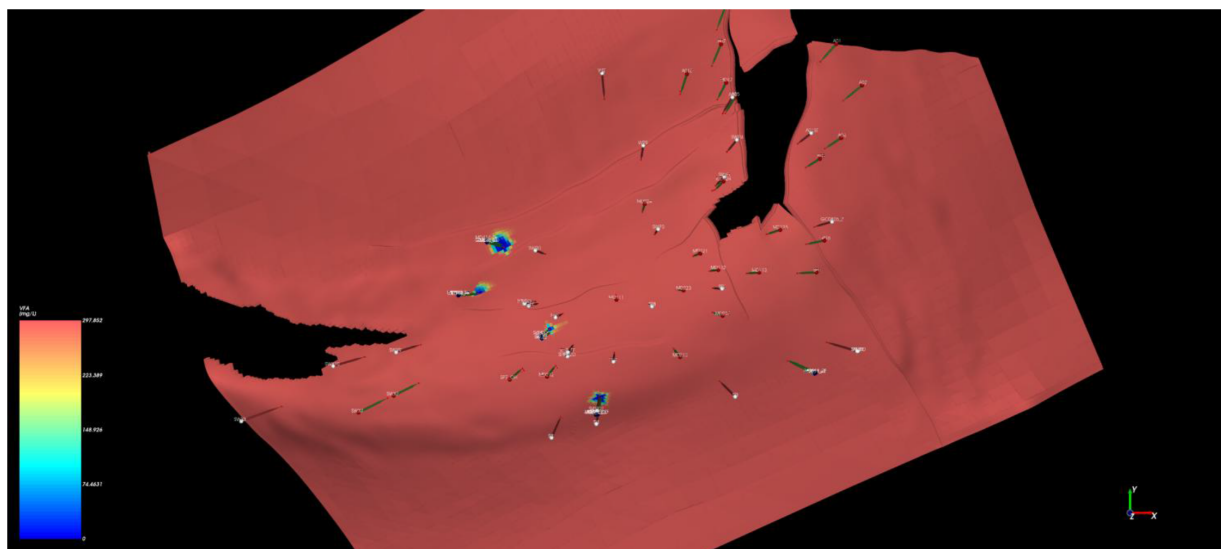


Figure 12—VFA distribution at the end of short-term prediction for the Reference scenario with 5X BoD LoSal sulphate level

The final line of argument in support of the short-term prediction results with no production of BSR H_2S can be demonstrated using thermal modeling comparison. Figs 13 and 14 show the temperature distribution in the reservoir at the beginning of LoSal water injection, as well as at the end of short-term prediction period.

It is very clear from Figs. 13 and 14 that there is hardly any change in temperature distribution from the start of LoSal water injection to the end of the short-term prediction period. The cooled zones are confined to the immediate vicinity of the injectors, hence greatly limiting the generation of H_2S . It is therefore no surprise that subdued generation of H_2S leads to no production of BSR H_2S during the short-term prediction period.

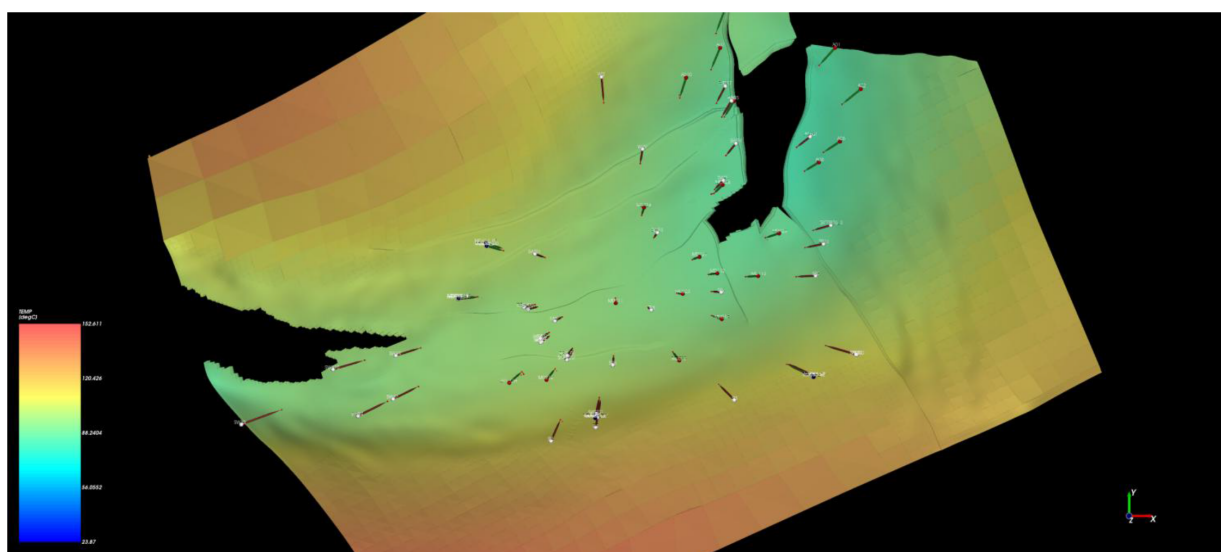


Figure 13—Reservoir temperature distribution at the start of LoSal water injection

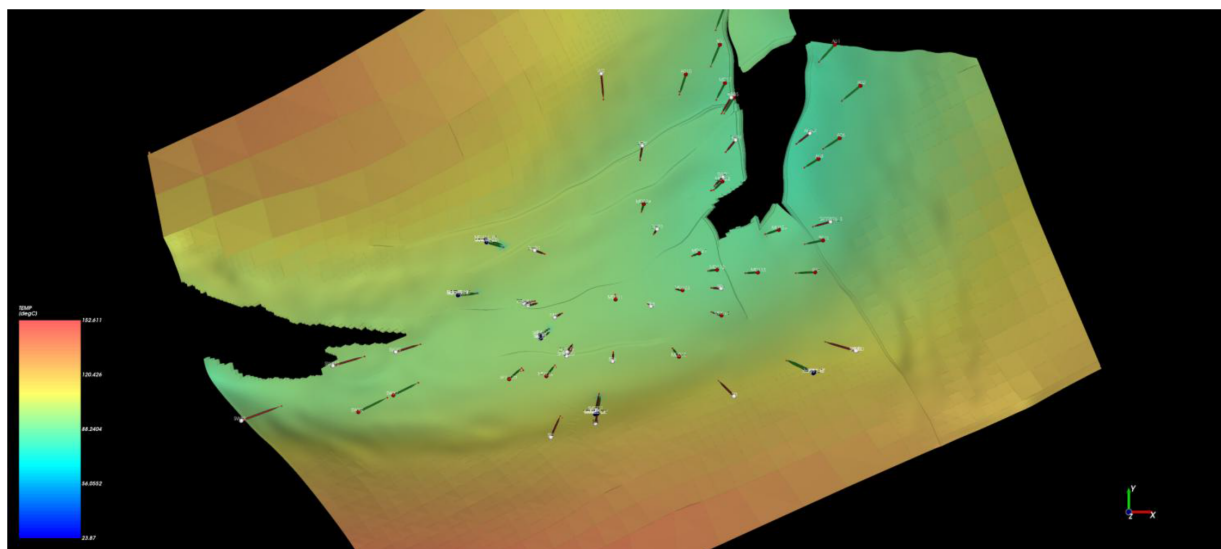


Figure 14—Reservoir temperature distribution for the Reference scenario at the end of the short-term prediction period

Long Term Prediction

A range of LoSal water sulphate level sensitivities have been run for the three reservoir souring scenarios. The topsides gas H_2S concentration profiles for these sensitivity runs are shown in Figs. 15 through 17. According to Fig. 15, any LoSal water sulphate levels below 7 mg/L will meet the topsides gas export H_2S specification during the life of the field for the Reference scenario. From Fig. 16, LoSal water sulphate levels below 5 mg/L will meet the topsides gas export H_2S specification during the life of the field for the High scenario. From Fig. 17, it is clear that LoSal sulphate levels as low as 3 mg/L will breach the gas export H_2S specification during the life of the field for the Low scenario. Since the Low scenario represents approximately the P90 case in the reservoir dynamic simulation ensemble models, the likelihood of breaching the topsides gas export H_2S specification during the life of the field is high. Therefore, for long-term production, mitigation measures (e.g., H_2S scavenging) should be in place to deal with the future breach of the topsides gas export H_2S specification.

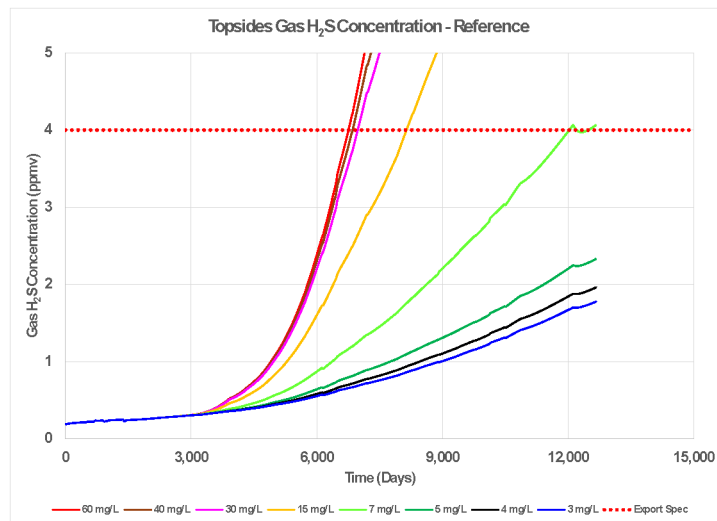


Figure 15—Topsides gas H_2S concentration profiles for the Reference scenario – different colors represent different sulphate levels in LoSal water

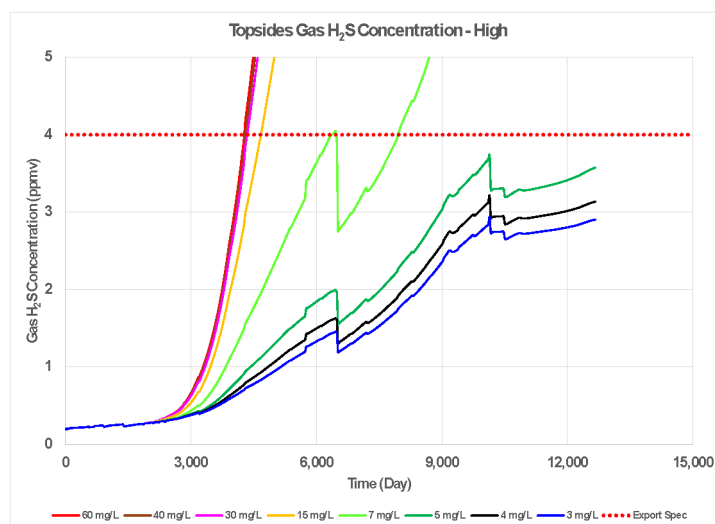


Figure 16—Topsides gas H₂S concentration profiles for the High scenario – different colors represent different sulphate levels in LoSal water

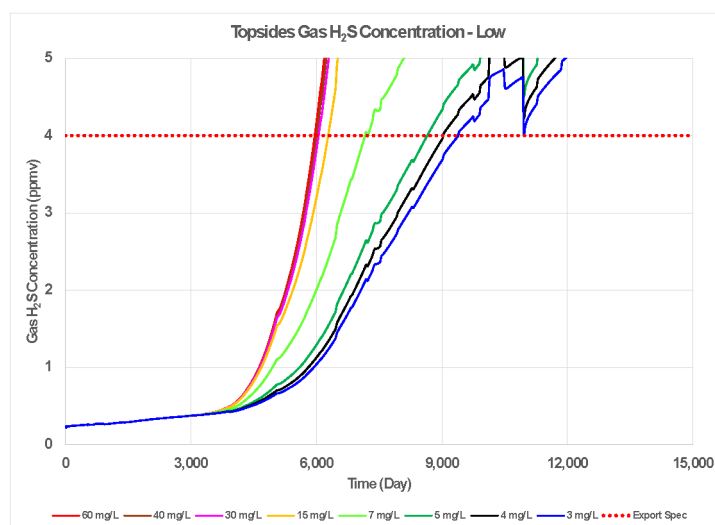


Figure 17—Topsides gas H₂S concentration profiles for the Low scenario – different colors represent different sulphate levels in LoSal water

Similar to the short-term prediction discussion, one can examine both passive tracer (VFA) and thermal modeling results to explain the reason why the long-term reservoir souring modeling predicts that H₂S production due to reservoir souring is inevitable during the life of the field.

The produced water VFA profiles for the three reservoir souring scenarios are shown in Fig. 18, in comparison with the VFA levels of the formation and LoSal injection water. It is evident from Fig. 18 that after about 5 years of LoSal water injection for the High scenario, the field is showing produced water VFA level coming down from the original formation water level, indicating injected water breakthrough. For the Reference and Low scenarios, the inferred injection water breakthrough happens after approximately 7 years after the start of LoSal water injection.

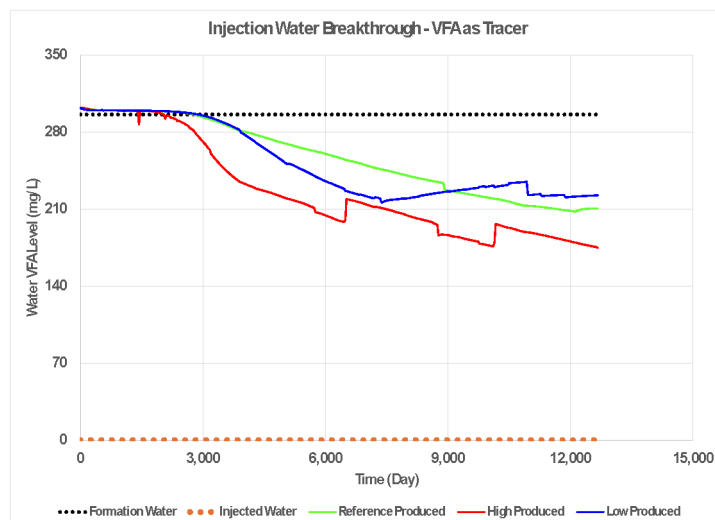


Figure 18—Produced water VFA profiles for the three reservoir souring scenarios in comparison with the formation and LoSal injection water

Figs. 11 and 19 together show the map comparison of Reference scenario VFA distribution for the long-term predictions, and Figs. 11 and 20 together show the map comparison of the High scenario VFA distribution for the long-term predictions. It is very clear from the two sets of comparison in Fig. 11 and Figs. 19 and 20 that, at the end of the long-term prediction run, VFA front as passive tracer has long arrived at some of the producers, indicating LoSal injection water breakthrough, which is necessary to the production of BSR H_2S . For thermal modeling, Figs. 13 and 21 together illustrate the map comparison of Reference scenario temperature distribution for the long-term predictions, and Figs. 13 and 22 together illustrate the map comparison of High scenario temperature distribution for the long-term predictions. Both sets of comparison demonstrate that, at the end of the long-term predictions, substantial cool zones around injectors have been established, even more so for the High scenarios than for the Reference scenario. These cool zones provide conducive environment in which BSR can be sustained and H_2S can be generated. All these sets of comparison demonstrate very clearly why for the long-term predictions, the chances of exceeding the topsides gas export limit are high, even without LoSal injection water sulphate level excursion.

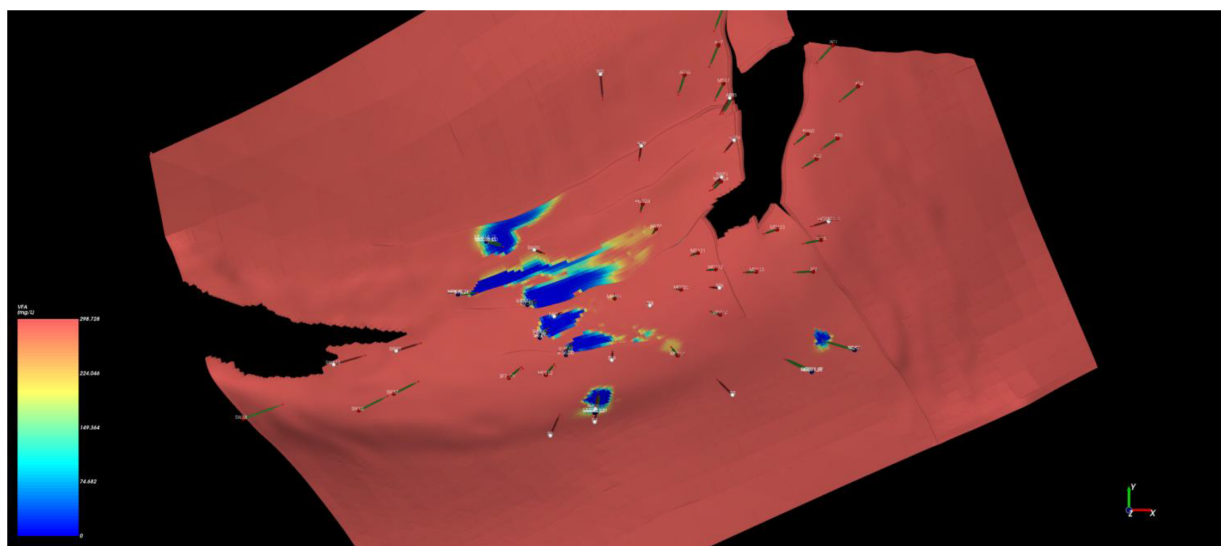


Figure 19—VFA distribution at the end of long-term prediction for the Reference scenario with 3X BoD LoSal sulphate level

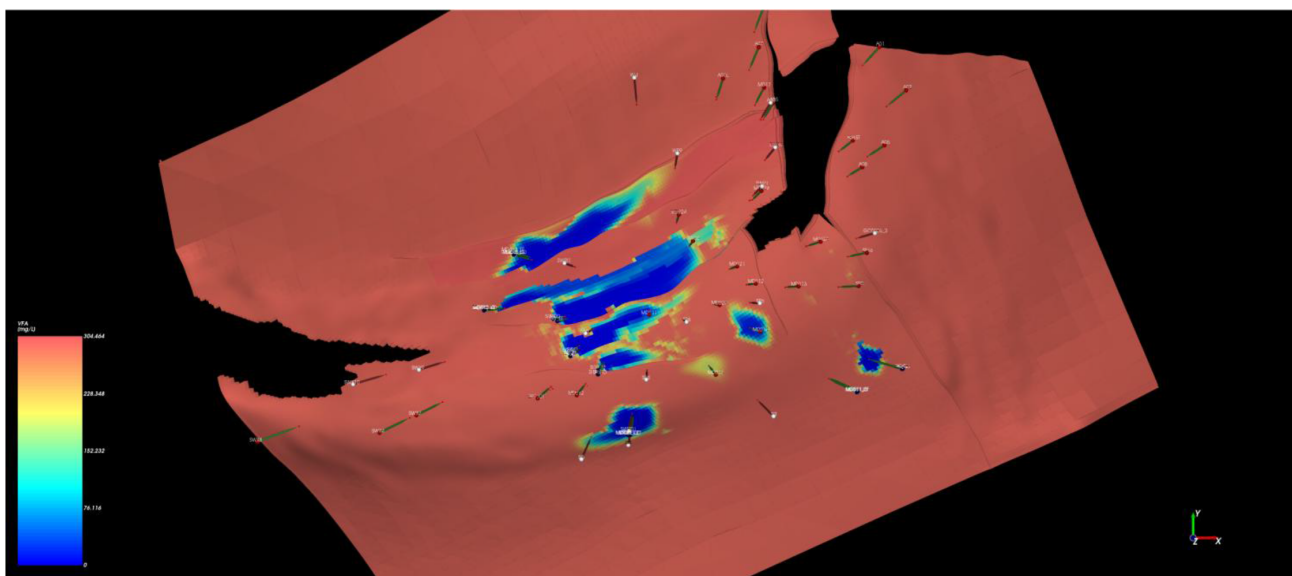


Figure 20—VFA distribution at the end of long-term prediction for the High scenario with 3X BoD LoSal sulphate level

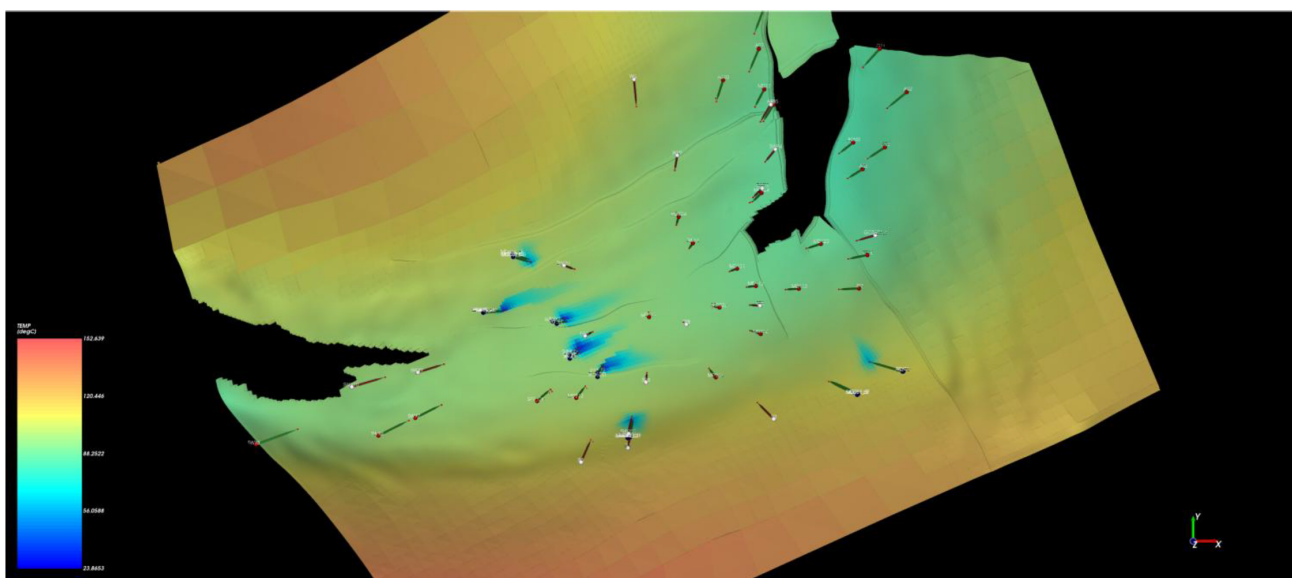


Figure 21—Reservoir temperature distribution for the Reference scenario at the end of the long-term prediction period

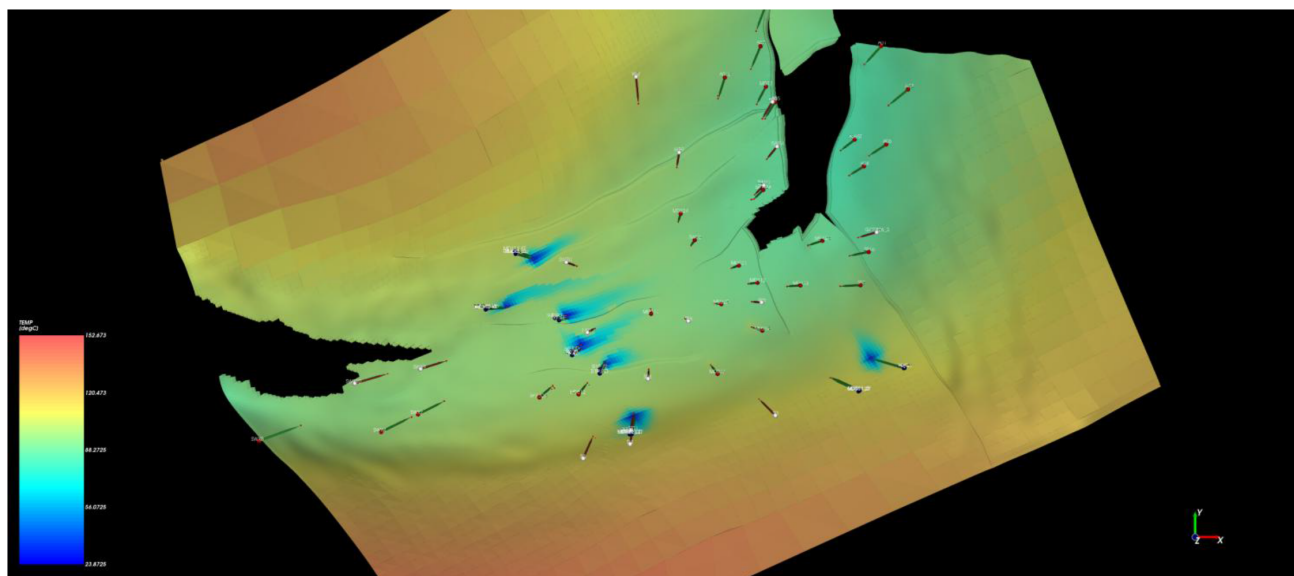


Figure 22—Reservoir temperature distribution for the High scenario at the end of the long-term prediction period

Conclusions and Recommendations

Based on the comprehensive reservoir dynamic, thermal and souring modeling study described above, one can conclude the following:

- For this field, reservoir dynamic performance ensemble models with probability distribution enable the establishment of three reservoir souring scenarios for reservoir souring modeling, Low, Reference and High, covering a wide range of water injection and production behavior.
- A conservative approach for reservoir souring modeling is adopted to assess the reservoir souring risks, as the stringent topsides gas export H_2S limit is strictly enforced by the third-party export line owner / operator. The conservative approach includes assumptions of aggressive biomass growth and H_2S generation once conditions are favorable; no reservoir H_2S scavenging capacity; no inhibitive benefit from high reservoir pressure; and the lowest LoSal injection water arrival temperature at sand-face.
- Temperature and sulphate supply are the more important controlling factors affecting the severity of reservoir souring for LoSal water injection.
- For highly undersaturated reservoirs such as this, with no free gas in subsurface throughout the life of the field, H_2S production will only occur after injection water breakthrough, even if conducive environment has been created for BSR (e.g., cool zones and sulphate supply)
- For the short-term topsides operations (within ~ 3 years of the start of LoSal water injection), the reservoir souring modeling has demonstrated that the chances of exceeding the topsides gas export H_2S limit are low, even for severe and sustained LoSal plant performance excursion (5X BoD).
- For the long-term management of topsides (out to 35 years from the start of LoSal water injection), however, the reservoir souring modeling results have demonstrated that the chances of exceeding the topsides gas export H_2S limit are high later in the field life, regardless of the sulphate level in the LoSal injection water. Hence, mitigation measures, such as H_2S scavenging, are recommended for managing the long-term risks of breaching the topsides gas export H_2S limit.

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